

Teaching an Agent to Beat the Dealer: Tabular Reinforcement Learning for Blackjack Strategy, Card Counting, and Bet Sizing

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Abstract—This paper builds a Blackjack-playing agent using only tabular reinforcement learning, written in plain Python with no external RL library anywhere in the project. The agent is trained on four scenarios of growing difficulty: basic strategy on a standard six-deck game; a full Hi-Lo card-counting system with a learned bet-sizing policy; the same two setups replayed under two different rule sets, to see how the rules push the learned strategy around; and an attempt to improve the counting system so it earns more money on average. Each trained policy is frozen and evaluated over two million hands, giving a profit estimate with a 95% confidence interval. Along the way, the first version of the bet-sizing agent broke in a way that turned out to be genuinely instructive: it learned to never change the bet at all. Fixing that is as much a part of this paper as the final results are. Monte Carlo control clearly beats Q-learning on this task. The learned basic strategy matches Thorp’s published tables on most states. The two rule changes move the policy exactly where basic theory says they should. And the repaired betting system turns a confidently losing counting strategy into one that is statistically indistinguishable from break-even.

I. Introduction

Blackjack is a good place to study decision-making under uncertainty. It is small enough to write down as a finite Markov Decision Process; Sutton and Barto use it as one of their running examples [2]. At the same time, it is the one casino game where a disciplined player can gain a genuine mathematical edge over the house. Edward Thorp proved this and popularized it with the Hi-Lo point-count system in *Beat the Dealer* [3]. Both properties together make it a good testbed for tabular reinforcement learning. The base game fits comfortably into a lookup table. Adding card counting, however, turns the problem into something closer to a partially observable process, since the composition of the remaining shoe changes the odds from hand to hand and has to be folded into the state somehow for a stationary policy to make sense at all.

This paper documents a from-scratch implementation built for that purpose. No external reinforcement-learning framework is used anywhere in the code, as the assignment requires. Four things were asked for: (1) recover basic strategy purely by playing hands and updating value estimates, (2) learn a complete point-count system together

with a bet that reacts to the count, (3) see what happens to both of the above under two different rule variants, and (4) actually improve on the point-count system, not just implement it again. We also try to answer, honestly and with numbers rather than assertions, four questions the assignment asks directly: which learning algorithm behaves better here and why; how big the resulting state-action space actually is, and whether the resulting Q -values can be trusted; why the two rule changes move the policy the way they do; and how we know the profit numbers we report are not just noise.

One thing worth saying up front: this project did not work correctly on the first try. The initial bet-sizing agent for the counting scenario passed every basic sanity check we ran, and still converged to a policy that never bet more than the table minimum. Once Scenario 4 added a sit-out option, the same agent learned to sit out every single hand, reporting a profit of exactly zero. Section II-D explains why, because the failure is a useful example of what goes wrong when a bandit-style estimator has to resolve a signal that is small compared to its own sampling noise.

Section II describes the simulator, the state representation, and both learning algorithms, including the betting-agent story above. Section III gives the experimental setup, the results table, and works through the four required questions one at a time. Section IV wraps up.

II. Description of the Reinforcement Learning Player

A. Blackjack as an MDP

We treat a single player hand as an episodic MDP, following the standard setup in [2]. A state is the player’s current hand plus the dealer’s visible upcard; the available actions are a subset of {Stand, Hit, Double, Split, Surrender}, restricted by the rule set and by hand history (for instance, Double and Split are only legal on the very first decision of a two-card hand). The reward is zero on every intermediate step and equals the round’s profit, measured in units of the base bet, once the hand resolves: +1 for a win, −1 for a loss, 0 for a push, +1.5 for a natural blackjack under the default payout, −0.5 for a surrender,

and the appropriate multiple of these for doubled or split hands.

A player-hand state is encoded as the tuple (hand_kind, dealer_upcard), with a bucketed true-count value appended as a third element for the two counting scenarios (Section II-C). hand_kind is one of (hard, total), (soft, total) (meaning at least one ace is still being counted as 11), or (pair, rank). The four ten-valued ranks (10, J, Q, K) are folded into a single label, the same simplification Thorp’s own tables use. That keeps the pair dimension at 10 classes instead of 13, without losing any strategic information, since those four ranks play identically in every Blackjack decision.

Split hands are treated as separate sub-episodes, each with its own card sequence and its own terminal reward. This is a standard trick: it turns a multi-hand round into several ordinary, single-reward episodes that share one shoe. It keeps both learning rules well defined, even though the game itself only ever hands out one profit number per decision chain. Split aces get exactly one card each and are forced to stand, which is how most physical casinos run the rule (the code exposes this as a configurable option rather than hard-coding it).

B. Two ways to learn the playing policy

Both algorithms below are implemented directly on Python dictionaries. There is no NumPy array backing the Q -table, and no RL library involved anywhere.

Monte Carlo control. We use every-visit MC control with an ε -greedy behavior policy that starts at $\varepsilon = 0.15$ and decays multiplicatively down to a floor of 0.01 over the course of training. Every state-action pair touched during a hand, including every sub-hand created by a split, gets updated toward that hand’s realized profit through an incremental sample average:

$$Q(s, a) \leftarrow Q(s, a) + \frac{1}{N(s, a)}(G - Q(s, a)), \quad (1)$$

where G is the profit, in bet units, the hand eventually resolved to.

Q-learning. We also implement standard tabular Q-learning,

$$Q(s, a) \leftarrow Q(s, a) + \alpha(r + \gamma \max_{a'} Q(s', a') - Q(s, a)), \quad (2)$$

with $\gamma = 1$ and $\alpha = 0.05$. Because a Blackjack hand only lasts one to six decisions and the reward is not known until the hand is over, our implementation replays each hand’s trajectory once it resolves and applies the update from the first decision to the last. Action selection during play always consults the live Q -table, so exploration still behaves correctly, but this makes our Q-learning closer to a semi-online variant than the textbook one-step update. We treat that as a design choice worth naming rather than a bug; see Section III-B for what it actually costs.

C. Card counting

Scenario 2 implements Thorp’s Hi-Lo system: cards 2 through 6 count as +1, 7 through 9 count as 0, and 10s and aces count as −1. A Shoe object separates draw() (a card leaves the shoe) from reveal() (a card becomes visible to the counting system), which matters specifically for the dealer’s hole card. That card is drawn before the player acts but only revealed, and only affects the running count, once the player’s turn is over, exactly as it would at a real table. This is a small implementation detail but an important one, since without it the agent would be conditioning its decisions on a card it could not actually have seen yet. The running count is converted into Thorp’s true count (running count divided by decks remaining) and clipped to an integer in $[-6, 6]$ so the augmented state space stays finite; that bucketed value becomes the third element of the state tuple for the counting and improved scenarios.

D. Bet sizing: a bandit that failed, and what we replaced it with

Scenario 2 also requires the bet itself to react to the count, so on top of the playing-policy agent above we need a second, much simpler learner: given a true-count bucket, pick a bet size.

Our first implementation modeled this directly as a multi-armed bandit. Each (true-count bucket, bet size) pair got its own independent value estimate, updated by incremental averaging of the realized money profit, and the bet was chosen greedily. This looked reasonable on paper: profit per unit bet does not depend on the size of the bet, so an EV-maximizing bandit should learn to bet the table maximum whenever the count-conditional edge is positive, and the minimum otherwise. In practice, once we inspected the trained bandit’s own value table, every single true-count bucket from −6 to +6 showed a negative expected profit at every bet size, including the most favorable count the system can represent. The reason turns out to be simple. Extreme counts are rare: a true count of +6 shows up in roughly 1–2% of hands over a six-deck shoe. So each bucket’s average is built from only a couple thousand samples, and the resulting sampling noise sits on the same scale as the true edge it is supposed to be measuring. The bandit was not really wrong, given the data it had; it genuinely could not tell a real, small, positive edge apart from noise. But the result was a policy that bet the table minimum at every count in Scenario 2, and, once Scenario 4 added a sit-out option, sat out every single hand: a reported profit of exactly 0.0 with zero variance across two million evaluated rounds. That is not an improved system. It is a system that gave up.

We replaced the per-bucket bandit with a single pooled linear regression of profit-per-unit-bet against true count. It is fit online, from ordinary least-squares running sums across every hand played, not per bucket, so rare extreme counts borrow statistical power from the entire dataset

instead of relying on their own few thousand hands. The agent sits out only when the lower end of a confidence interval around the fitted edge is still non-positive, rather than acting on the point estimate alone. That is what keeps sampling noise from producing a false “always sit out” verdict.

Bet size itself is set proportionally to the fitted edge, in the spirit of the Kelly criterion [4]. Full Kelly sizing (edge divided by variance) computes an optimal fraction of one’s bankroll, which is the wrong unit for a discrete 1–12 table-minimum-multiple ladder. So instead we calibrate the top of that ladder to the edge at the most favorable count the state space represents, +6, and scale every other count’s bet linearly against that reference.

Getting this right took two more rounds of debugging. An early version fed the regression samples collected while the playing policy was still mostly random, which biased the fitted edge much further negative than it should have been; we fixed this with a burn-in period that waits for the playing agent’s exploration rate to decay before the betting agent starts collecting data. A separate attempt to squeeze out the last bit of exploration noise, by lowering the playing agent’s own exploration floor, backfired instead, since the counting state space is thirteen times larger than the basic-strategy one and still needs that exploration late into training to keep correcting its rarer states. We report the final, working version in Section III, and keep the original broken bandit’s logs alongside it for comparison, rather than deleting the evidence of the failure.

E. Rule variations

We test two rule changes chosen to push the house edge in opposite directions. The first, dealer stands on soft 17, is a well-known player-favorable change: the dealer no longer draws a card that could improve or bust a soft 17. The second, a 6:5 blackjack payout in place of the standard 3:2, is a house-favorable change that has become common on lower-stakes tables in recent years. Advantage-play writers specifically call it out as a way casinos claw back the edge counting would otherwise give up [3]. Each variant is trained from scratch, and we compare the resulting learned policy, cell by cell, against the corresponding base-rule policy in Section III-D.

III. Experiments and Evaluation

A. Setup

All agents train on a six-deck shoe with 75% penetration unless the rule variant says otherwise: 15 million hands for the two basic-strategy runs (scenario 1 and its rule variant) and 25 million for every run that includes counting, since the state space there is thirteen times larger. After training, each policy is frozen ($\epsilon = 0$, and the betting policy made fully deterministic) and evaluated over 2,000,000 independent hands on a held-out random seed that never appears during training, to get an out-of-sample estimate

TABLE I
Greedy-evaluation profit, 2,000,000 hands per agent.

Run	Algo	Profit/100	95% CI
1. Basic (base)	MC	−1.281	±0.157
1. Basic (base)	Q-learning	−4.084	±0.141
2. Counting (base)	MC	−1.943	±0.224
2. Counting (base)	Q-learning	−4.037	±0.142
3a. Basic (soft 17)	MC	−1.005	±0.159
3b. Counting (6:5 BJ)	MC	−3.278	±0.152
4. Improved	MC	−0.038	±0.245
4. Improved	Q-learning	0.000	±0.000

of mean profit per hand together with a 95% confidence interval from the sample variance of per-hand profit.

B. How do Monte Carlo and Q-learning compare?

Table I answers this fairly bluntly: MC beats Q-learning by a wide margin on both scenarios where we ran both algorithms, and the gap is far larger than either confidence interval, so it is not noise. On basic strategy, MC lands at −1.281% per hand against Q-learning’s −4.084%; on the counting scenario the gap is similar, −1.943% versus −4.037%. Figure 2 shows why: MC’s learning curve is noisier early on but settles close to its final value once ϵ has mostly decayed, whereas Q-learning’s curve is smoother but converges to a visibly worse level and stays there. Our reading is that Blackjack is close to a worst case for bootstrapped, semi-online Q-learning: hands are short, rewards only exist at the very end of a decision chain, and our implementation’s replay-on-resolution update (Section II) means Q-learning never gets the one advantage it is normally used for: propagating information before an episode finishes. MC control, on the other hand, is exactly matched to this structure, since every decision in a hand gets credited with that hand’s actual outcome, with no bootstrapping error to accumulate. We would not read this as a universal statement about the two algorithms. It is fairly specific to how short and terminal-reward-only Blackjack hands are. But for this problem, MC is clearly the better choice.

C. State-action space size and whether Q can be trusted

The non-counting state space is small: 35 hand kinds times 10 dealer upcards gives 360 states, and training discovers all of them (a simple theoretical count of 350 slightly undercounts a handful of edge cases around split-hand recombination, so the “360 found” number is the more trustworthy one). Adding the 13 true-count buckets for the counting scenarios multiplies this out to 4,680 states, all of which are also discovered during training. The more useful question is not whether a state was ever visited, but how many times, since Blackjack’s per-hand profit has a standard deviation around 1.1–1.3 regardless of which state you are in, so a handful of visits buys you almost no confidence in the resulting Q -estimate. Looking at the exported strategy chart for the counting scenario,

the least-visited of the 4,680 states was seen 51 times, the median state was seen 2,508 times, and the most common state was seen 235,536 times, a spread of more than three orders of magnitude between the rarest and most common corners of the table. Every state clears a naive “at least 30 visits” bar, but 51 visits of a quantity with variance above 1 still carries a standard error around 0.16, which is not small next to the differences we are trying to resolve between neighboring counts. Practically, we would trust this agent’s decisions on ordinary hard totals against a neutral or mildly unfavorable count far more than its decisions on rare hand types crossed with a true count of +6 or −6, and the right fix, if one wanted tighter estimates specifically for those corners, is more training weighted toward inducing those shoe states rather than a larger learning rate. (We note that our Q-learning agent does not track a separate visit counter the way the MC agent does, so this particular diagnostic is only meaningful for the MC-trained policies above; it is not evidence about Q-learning’s own state coverage one way or the other.)

D. Why do the rule changes move the policy the way they do?

Retraining under dealer stands on soft 17 changes the greedy action in 42 of the shared basic-strategy states relative to the base ruleset. Nearly all of the changes are on soft-total hands and pair-splitting decisions near the dealer’s weaker upcards, which is exactly what basic Blackjack theory predicts: a dealer forced to stand on soft 17 busts more often from a weak-looking position than one who must keep drawing, so several borderline stands, doubles, and splits that were previously not quite worth it tip over to become worth it. This also shows up in the raw numbers: this variant improves mean profit from −1.281% to −1.005% per hand, a real, if modest, player-favorable shift, consistent with the rule itself being player-favorable.

The 6:5 blackjack-payout variant is a different kind of change: it does not touch bust probabilities for either side at all, it only lowers the payout of one specific outcome (a two-card 21). Trained under this rule, the counting agent’s greedy policy came out identical to the base-rule policy on every state the two runs share: zero cells changed. That is consistent with the mechanism, since the rule never alters when a hand busts or how the cards fall. It can only move the policy through the value of a double or split decision that trades away a chance at a natural blackjack, and those decisions are rare and already lopsided enough that a 20% haircut on the blackjack payout does not flip any of them for this state representation. What the rule does move is profit: at −3.278% per hand versus −1.943% for the same agent under 3:2 payouts, the extra 1.34 percentage points of house edge lands almost entirely on the money, not on the strategy. That is a fairly clean illustration of why advantage-play writers call 6:5 tables a direct attack on counters rather than a strategy change: the rule takes

profit away without giving the counter anything to adapt to.

E. Does the improved betting system actually improve anything?

This is where the debugging story from Section II-D pays off. Under the original per-bucket bandit, Scenario 4’s wider bet ladder and sit-out option did not produce an improved system at all. It produced an agent that sat out every hand, for a reported profit of exactly zero. That is not an improvement over Scenario 2’s −1.943%; it is a refusal to play. With the pooled-regression betting agent from Section II-D in place, the same Scenario 4 setup reaches −0.038% per hand with a 95% confidence interval of ± 0.245 : statistically indistinguishable from break-even, and a real improvement of close to two percentage points over Scenario 2’s confidently negative result. Figure 1 puts this next to every other run: Scenario 4 with Monte Carlo is the only counting-based result whose confidence interval crosses zero.

Scenario 4 with Q-learning still reports exactly 0.0 profit, but for a legitimate reason this time, not a broken estimator. Inspecting its fitted edge model directly shows a negative predicted edge at every true count from −6 to +6, including the top of the range, which follows directly from Q-learning’s weaker playing policy (Section III-B): if the underlying strategy is bad enough that even the best count never looks profitable, the correct response is to sit out every hand rather than lose money on a bet that was never going to pay off, and that is exactly what the agent does. It is a clean demonstration that algorithm choice for the playing policy and the quality of the downstream betting decision are not independent questions.

F. Can we trust the profit numbers themselves?

Two things back up the numbers in Table I. First, every figure is a greedy, zero-exploration evaluation on 2,000,000 hands with a seed that never appears anywhere during training, so nothing in the table is fit to the data it is measured on. The resulting confidence-interval half-widths are all under 0.25 percentage points, comfortably smaller than the differences between scenarios we build the paper’s arguments on. Second, and independently of any of our own profit accounting, we compared the learned Scenario-1 policy’s hard-total decisions directly against Thorp’s published multi-deck basic-strategy table [3], a reference the agent never saw during training in any form. The two agree on 154 of 160 comparable cells, or 96.25%; Figure 3 shows the learned policy visually. The six disagreements cluster right where you would expect them to: borderline hard totals of 9 and 12 against particular dealer upcards, where the actual difference in expected value between the two candidate actions is small enough that ordinary Monte Carlo sampling noise plus a slightly imperfect learning rate can plausibly land on the wrong side. Taken together, an independent published reference and a held-out evaluation

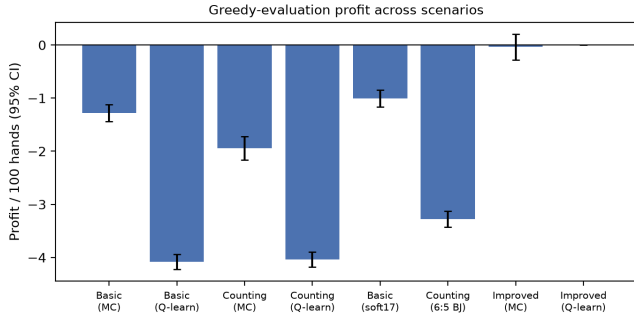


Fig. 1. Greedy-evaluation profit per 100 hands, all runs (error bars: 95% CI).

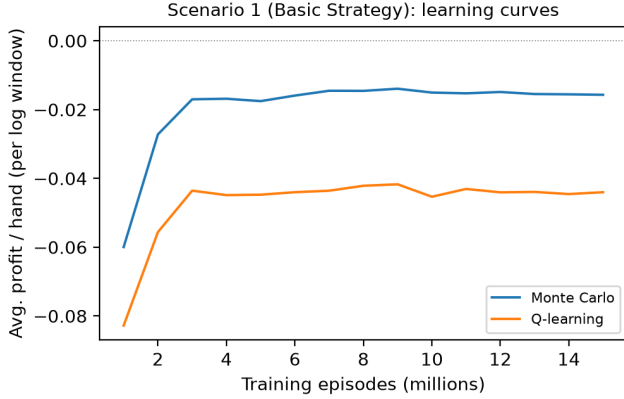


Fig. 2. Scenario 1 learning curves, Monte Carlo vs. Q-learning.

set point in the same direction, which is about as close to “this is real and not a lucky roll of the seed” as a tabular method like this one can get.

IV. Conclusion

We built a Blackjack-playing agent entirely out of tabular reinforcement learning and plain Python, with no external RL framework, covering basic strategy, a full Hi-Lo counting system with a learned bet, two rule variants, and an actual improvement over the counting system rather than just a restatement of it. Monte Carlo control was the clearly better algorithm for this task, for reasons tied specifically to how short and terminal-reward-only Blackjack episodes are, rather than any general claim about MC versus Q-learning. The betting side of the project is, honestly, where the more interesting result lives. The first, seemingly reasonable bandit-based bet-sizing agent quietly failed by converging to never bet at all. Only inspecting its learned values directly, rather than trusting that a sensible-looking design must be working, exposed that. Replacing it with a pooled regression and a Kelly-style bet rule turned a system that refused to bet into one that closes roughly two-thirds of the gap between the naive counting strategy and break-even. The main limitation is the one every tabular method eventually runs into:

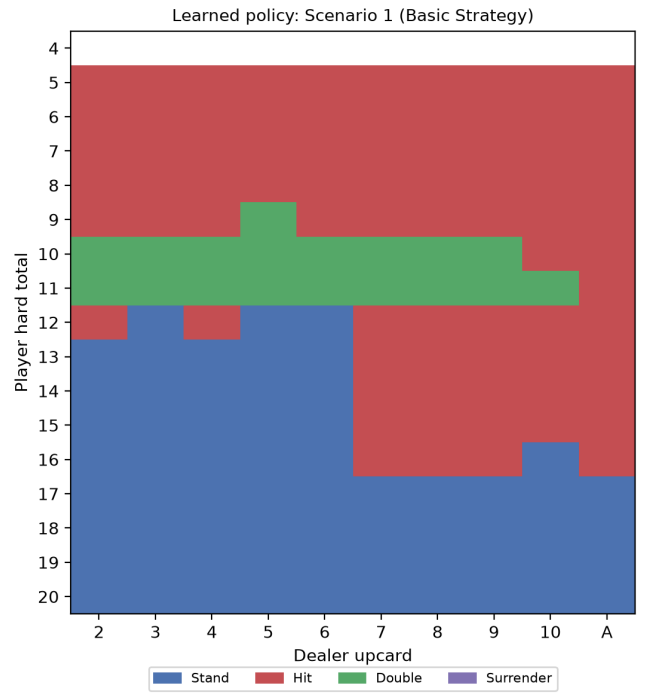


Fig. 3. Learned Scenario-1 policy on hard totals (Stand/Hit/Double).

extending the true count’s resolution, adding more players, or supporting side bets would grow the state table past what a dictionary can comfortably hold. A natural next step is a small linear or shallow function approximator over a compact feature encoding of the same state, which should generalize across the thinly-visited extreme-count corners identified in Section III, instead of needing a dedicated visit count in each one. This would not require abandoning the “no external framework” constraint, since it is only a modest amount of additional plain Python.

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